LOXIN INJECTION (5 ML AMP)

DESCRIPTION:
LOXIN INJECTION (5 ML AMP): A clear, colourless or pale yellow liquid.

COMPOSITION: Each ampoule contains Dopamine Hydrochloride 40 mg/ml. Total contents of each ampoule is 200 mg Dopamine Hydrochloride.

PHARMACODYNAMICS:
Dopamine Hydrochloride exerts an inotropic effect on the myocardium resulting in an increased cardiac output. Dopamine Hydrochloride produces less increase in myocardial oxygen consumption than isoproterenol and its use is usually not associated with a tachyarrhythmia.
Dopamine hydrochloride usually increases systolic and pulse pressure with either no effect or a slight increase in diastolic pressure. Blood flow to peripheral vascular beds may decrease while mesenteric flow increases. Dopamine hydrochloride has also been reported to dilate the renal vasculature presumptively by activation of a "dopaminergic" receptor. This action is accompanied by increases in glomerular filtration rate, renal blood flow and sodium excretion. An increase in urinary output produced by dopamine is usually not associated with a decrease in osmolality of the urine.

PHARMACOKINETICS:
Distribution, Metabolism, Excretion: It has been shown in animals that the distribution of dopamine is to all tissues of the body except the central nervous system. In man it is metabolised in the liver and to a lesser extent in the blood by monoamine oxidase and catechol-o-methyl transferase. Dopamine is excreted by the kidneys as its metabolites. In the presence of normal renal function, its excretion is rapid and virtually complete.
Following intravenous administration, onset of action is 5 minutes with a duration of action of less than 10 minutes (in patients receiving monoamine oxidase inhibitors this may be as long as 1 hour). The plasma half life of an intravenous bolus is about 2 minutes.
Dopamine also inhibits alpha and beta adrenergic effects which become more pronounced with dosage increases.

INDICATIONS:
Dopamine hydrochloride is indicated for the correction of haemodynamic imbalances present in the shock syndrome due to myocardial infarctions, trauma, endotoxic septicaemia, open heart surgery, renal failure and chronic cardiac decompensation as in congestive failure.

Poor perfusion of vital organ: Urine flow appears to be one of the better diagnostic signs by which adequacy of vital organ perfusion can be monitored.
Nevertheless, the physician should also observe the patient for signs of reversal of confusion or comatose condition. In a number of oliguric or anuric patients, administration of dopamine hydrochloride has resulted in an increase in urine flow which in some cases reached normal levels. Dopamine hydrochloride may also increase urine flow in patients whose output is within normal limits and thus may be of value in reducing the degree of pre-existing fluid accumulation. It should be noted that at doses above those optimal for the individual patient, urine flow may decrease, necessitating reduction of dosage. Concurrent administration of dopamine hydrochloride and diuretic agents may produce an additive or potentiating effect.

Low cardiac output: Increased cardiac output is related to the direct inotropic effect on the myocardium by dopamine hydrochloride. Increased cardiac output at low or moderate doses appears to be related to a favourable prognosis. In many instances, the renal fraction of the total cardiac output has been found to increase. Increase in cardiac output produced by dopamine hydrochloride is not associated with substantial decreases in systemic vascular resistance (SVR) as may occur with isoprenaline.

Hypotension: Hypotension due to inadequate cardiac output can be managed by administration of low to moderate doses of dopamine hydrochloride which have little effect on SVR. At high therapeutic doses, the alpha adrenergic activity of dopamine hydrochloride becomes more prominent and thus may correct hypotension due to diminished SVR. It is suggested that the physician administer dopamine hydrochloride as soon as a definite trend toward decreased systolic and diastolic pressure becomes evident.

RECOMMENDED DOSAGE:
Warning: This is a potent drug. It must be diluted before administration to patient.

Suggested dilution: Add 5 mL containing 200 mg Dopamine hydrochloride injection, to either a 250 mL or 500 mL I.V. container of a suitable intravenous solution. Sodium Chloride injection or Dextrose 5% injection have been found suitable. These solutions will yield a final concentration for administration as follows:
250 mL dilution contains 800 mcg/mL of dopamine hydrochloride.
500 mL dilution contains 400 mcg/mL of dopamine hydrochloride.
Dilution of dopamine hydrochloride should be made just prior to administration.

Rate of administration: Dopamine hydrochloride, after dilution, is administered intravenously through a suitable intravenous catheter or needle. An I.V. drip chamber or other suitable metering device is essential for controlling the rate of flow in drops/minute.
Treatment of all patients requires constant evaluation of therapy in terms of the blood volume, augmentation of myocardial contractility, and distribution of peripheral perfusion. Dosage of dopamine hydrochloride should be adjusted according to the patient's response with particular attention to diminution of established urine flow rate, increasing tachycardia or development of new dysrhythmias as indices for decreasing or temporarily suspending the dosage. As with all potent intravenous administered drugs, care should be taken to control the rate of administration so as to avoid inadvertent administration of a bolus of drug.

Administration and Adult Dosage, IV for shock, by infusion only (in any non-alkaline I.V. fluid): 2-5 µg /kg bodyweight/min initially, increasing gradually in increments of 5-10 µg/kg/min up to 20-50 µg/kg/min, titrating dosage to desired response in each patient. Most patients can be maintained on 20 µg/kg/min or less. Dosages over 50 µg/kg/min should be used only with careful monitoring of hemodynamic parameters. I.V. for chronic refractory congestive heart failure 0.5 µg/kg/min initially, increasing gradually until increases in urine flow, diastolic blood pressure or heart rate is observed. Most patients respond to 1-3 µg/kg/min.
Care must be taken in patients with cardiac decompensation to avoid alpha-adrenoreceptor induced vasoconstriction and increased afterload. These patients should be started on a dose of 1-2 mcg/kg/minute and the rate of infusion increased with caution. Patients with occlusive vascular disease should also be commenced on a similar low dose.

Elderly patients: Should be monitored for signs of peripheral ischaemia.

Paediatric dosage: Safety and efficacy not established.

ROUTE OF ADMINISTRATION : For i.v infusion after dilution

CONTRAINDICATIONS:
Dopamine hydrochloride should not be used in patients with:
1. Phaeochromocytoma: Dopamine, as a catecholamine, will have an additive effect to an already abnormally high catecholamine level in the presence of the tumor.
2. Atrial or ventricular tachyarrhythmias.
3. Cyclopropane and halogenated hydrocarbon anaesthetics should be avoided (see Drug Interactions).
4. Hyperthyroidism.

WARNINGS AND PRECAUTIONS:
Use in children: The safety and efficacy of this drug in children has not been established.

Precautions:
Avoid Hypovolaemia: Prior to treatment with dopamine hydrochloride, hypovolaemia should be fully corrected, if possible, with either whole blood or plasma as indicated.

Decreased pulse pressure: If a disproportionate rise in the distolic pressure (i.e. a marked decrease in the pulse pressure) is observed in patients receiving dopamine hydrochloride, the infusion rate should be decreased and the patient observed carefully for further evidence of predominant vasoconstrictor activity, unless such an effect is desired.

Extravasation: Dopamine hydrochloride should be infused into a large vein whenever possible to prevent the possibility of extravasation into tissue adjacent to the infusion site. Extravasation may cause necrosis and sloughing of surrounding tissue. The infusion site should be continuously monitored for free flow.

Occlusive vascular disease: Patients with a history of occlusive vascular disease (for example arteriosclerosis, arterial embolism, Raynaud's disease, cold injury, diabetic endarteritis and Buerger's disease) should be closely monitored for any changes in color or temperature of the skin in the extremities. If a change in skin color or temperature occurs and is thought to be the result of compromised circulation to the extremities, the benefits of continued dopamine hydrochloride infusion should be weighed against the risk of possible necrosis. This condition may be reversed by either decreasing or discontinuing the rate of infusion.

Pulmonary Hypertension: Pulmonary hypertension may be exacerbated due to dopamine induced pulmonary vasoconstriction. Where this has occurred, isoprenaline must be considered as an alternative inotropic agent.

Impaired renal & hepatic function: As the effect of dopamine on impaired renal and hepatic function is not known, close monitoring is advised.

IMPORTANT antidote for peripheral ischaemia: To prevent sloughing and necrosis in ischaemic areas, the area should be infiltrated as soon as possible with 10 to 15 mL of saline solution containing from 5 to 10 mg of phentolamine, an adrenergic blocking agent. A syringe with a fine hypodermic needle should be used, and the solution liberally infiltrated throughout the ischaemic area. Phentolamine should be given as soon as possible after the extravasation is noted.

LOXIN INJECTION contains sodium metabisulphite, a sulphite that may cause allergic-like reactions, including anaphylactic symptoms and life-threatening or less severe asthmatic episodes in certain susceptible people. The overall prevalence of sulphite sensitivity in the general population is unknown and probably low. Sulphite sensitivity is seen more frequently in asthmatic than non-asthmatic people. Do not use if known to be hypersensitive to bisulphites.

INTERACTIONS WITH OTHERS MEDICAMENTS:

Drug Interactions:

a) Dopamine is metabolised by MAO, and inhibition of this enzyme prolongs and potentates the effect of dopamine hydrochloride.
b) Patients who have been treated with monoamine oxidase (MAO) inhibitors prior to the administration of dopamine hydrochloride will require substantially reduced dosage. The starting dose in such patients should be reduced to at least one-tenth (1/10) of the usual dose.
c) Avoid cyclopropane or halogenated hydrocarbon anaesthetics; cyclopropane or halogenated anaesthetics increase cardiac autonomic irritability and therefore may sensitize the myocardium to the action of certain intravenously administered catecholamines.
d) Dopamine hydrochloride should be used with EXTREME CAUTION in patients inhaling cyclopropane or halogenated hydrocarbon anaesthetics, because of the theoretical arrhythmogenic potential.

The cardiovascular effects of dopamine are antagonized by both alpha and beta blocking drugs, the former blocking the peripheral vasoconstriction produced by dopamine and the latter antagonizing the cardiac effects. Hypotension may be observed with concurrent use of vasodilators such as nitroglycerin, nitroprusside and calcium channel blockers.
Hypotension, bradycardia and possibly cardiac arrest have been observed in patients receiving phenytoin. If phenytoin must be administered, carefully monitor the patient's blood pressure and discontinue if hypotension occurs.
Dopamine may increase the effect of diuretic agents.

Concurrent use of:

- Digitalis glycosides may increase the risk of cardiac arrhythmias. Caution and close ECG monitoring are very important if concurrent use is necessary.
- Ergotamine may produce peripheral vascular ischaemia and gangrene and is not recommended.
- Ergot alkaloids or oxytocin may potentates the pressor effect of dopamine with possible severe hypertension and rupture of cerebral blood vessels.
- Ergot alkaloids should also be avoided because of the possibility of excessive vasoconstriction. Tricyclic antidepressants and guanethidine may potentates the pressor response to dopamine.

USE IN PREGNANCY AND LACTATION:

Use in Pregnancy: The use of any drug in pregnant women or women of childbearing age requires that the expected benefit of the drug be carefully weighed against the possible risk to the mother and the child. It is not known whether dopamine hydrochloride crosses the placenta barrier.
Use in Lactation: It is not known whether dopamine enters breast milk. It is inactive when ingested orally but, nonetheless, it is not recommended for nursing mothers unless the expected benefits outweigh any potential risks.

SIDE EFFECTS:

Adverse reactions have been observed in 19% of patients during clinical evaluation, however, only half of these are attributed to dopamine.
Treatment was terminated in 5% of all patients due to adverse reactions. The most frequent adverse reactions to dopamine hydrochloride included ectopic beats, nausea, vomiting, tachycardia, anginal pain, palpitation, dyspnoea, headache, hypotension and vasoconstriction. Other adverse reactions which have been reported infrequently were aberrant conduction, bradycardia, piloerection, widened QRS complex, azotaemia and elevated blood pressure. Pedal gangrene has occurred in a few patients with pre-existing vascular disease. (see Precautions). Fatal ventricular arrhythmias have been reported on rare occasions.

SYMPTOMS AND TREATMENT OF OVERDOSE:

Overdosage: In case of accidental overdosage, as evidenced by excessive blood pressure elevation, reduce rate of administration or temporarily discontinue dopamine hydrochloride until patient's condition stabilizes. Since the duration of action of dopamine hydrochloride is quite short, no additional remedial measures are usually necessary. If these measures fail to stabilize the patient's condition, use of the short acting alpha adrenergic blocking agent, phentolamine should be considered.

INCOMPATIBILITIES:

Do not add dopamine hydrochloride to any alkaline solution since it is inactivated in alkaline solution. Dopamine is also sensitive to oxidizing agents and iron salts.

STORAGE CONDITIONS:

Store below 30°C. Protect from light and freezing.
Keep out of reach of children. *Jauhkan daripada kanak-kanak.*

DO NOT USE THE INJECTION IF IT IS DARKER THAN SLIGHTLY YELLOW OR DISCOLOURED IN ANY OTHER WAY. MUST BE DILUTED WITH A SUITABLE PARENTERAL VEHICLE PRIOR TO I.V. INFUSION.

Shelf Life of Product after Dilution: Injection should be diluted immediately prior to administration. After dilution in appropriate intravenous solution for infusion, dopamine is stable for at least 24 hours.

PACK SIZES: Pack in 10's ampoules of 5 ml / box.

SHELF LIFE: Please refer to outer package.

PRODUCT REGISTRATION HOLDER AND MANUFACTURER:

DUOPHARMA (M) SDN BHD
Lot 2599, Jalan Seruling 59, Kawasan 3,
Taman Klang Jaya, 41200 Klang, Selangor, MALAYSIA